

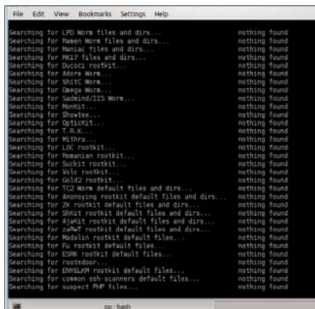


Figure 1: Root-kit scanning

Table 1: Chkrootkit components

Chkrootkit Program	Purpose
Chkrootkit	Main script to check for tampered system files
strings.c	Detects and performs string replacement
ifpromisc.c	Checks network interface for promiscuous mode
chklastlog.c, chkwtmip.c	Checks if lastlog and wtmp entries are deleted
chkproc.c, chkdirs.c	Checks for Linux kernel module-based Trojans

Installation

Chkrootkit installation is straightforward on Red Hat and a few other distributions. You can download the most recent tarball from www.chkrootkit.org/download to a temporary folder. It is recommended that you perform an MD5 check, and then decompress the tarball with `tar xvfz chkrootkit.tar.gz`. Change to the extracted directory, and compile it with `make sense`.

For Debian Linux flavours such as Ubuntu, simply use `sudo apt-get install Chkrootkit`.

Usage

The very first step recommended after installing Chkrootkit is to run `ifpromisc`. This checks all network interfaces if in promiscuous mode, which should not be the case unless the system is infected with a root-kit prior to installation. Once this check is done, the next step is to run the tool without any command-line options. Figure 1 shows how a typical Ubuntu system is scanned and checked for various supported anomalies and root-kits. Figure 2 shows how Chkrootkit checks all kernel processes and system files, based on an internally stored checklist. If any common OS utility

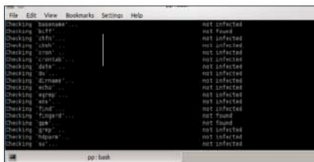


Figure 2: Checking all kernel processes and system files



Figure 3: Report of an infected file

(such as *fingerd*, as seen in Figure 2) is not detected, it reports this, so that administrative scripts can check whether a file that should normally be present has been deleted.

Though running Chkrootkit without any command-line options is usually sufficient, systems administrators may want more flexibility to script it. Table 2 describes a few command options with examples.

Table 2: Chkrootkit command-line options

Option	Explanation	Example
-l	Lists the tests supported	<code>>chkrootkit -l</code>
[testname]	Scans for a specific test	Check Sniffer command for "ps" Trojan <code>>chkrootkit ps sniffer</code>
-x	Expert mode (displays each action taken on each file scanned)	<code>>chkrootkit -x more</code>
-q	Quiet mode. Only displays if a binary is found to be 'Infected'	<code>>chkrootkit -q</code>
-r	Uses specified <i>dir</i> as root <i>dir</i> . Useful in scanning a suspicious machine from a healthy one. Also useful in scanning mounted volumes.	Scans a volume mounted under "mnt1" <code>>chkrootkit -r /mnt1</code>

It is also important to understand how Chkrootkit displays output. Typically, phrases such as 'Not found' or 'Not infected' are displayed. When a root-kit is found, or if the presence of a root-kit is suspected, the output highlights it with 'INFECTED', or "The following suspicious files or directories have been found." Please refer to Figure 3, which